# REAL-TIME LIGHTWEIGHT SEMANTIC ROAD IMAGE SEGMENTATION FOR AUTONOMOUS DRIVING ENVIRONMENTS: FOCUSING ON MOBILENETV2-BASED U-NET

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ABSTRACT

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***K*eywords** Semantic Segmentation · Autonomous Driving · MobileNetV2 · U-Net · Real-Time Processing · Edge Computing

## 1 Introduction

Accurate perception of the surrounding environment is the core of ensuring safety in autonomous driving technology. In particular, semantic segmentation, which understands scenes at the pixel level, clearly distinguishes drivable areas, pedestrians, and other elements, enabling safe path planning. However, autonomous vehicles must perform real-time processing on on-board edge devices without network latency. Existing high-performance deep learning models have massive FLOPs and parameters, presenting physical limitations for deployment in mobile and edge environments. To address this, this study proposes a hybrid architecture that combines the U-Net structure [1]—known for its excellent spatial information recovery—with an extremely lightweight MobileNetV2 [2] backbone as the encoder, allowing it to operate in real-time even under limited hardware resources.

## 2 Related Works

Early models like FCN (Fully Convolutional Network), DeepLab series, and PSPNet achieved high accuracy (mIoU) in road environment segmentation but utilized heavy backbones (e.g., ResNet-101), making it difficult to satisfy the real-time processing constraints (e.g., over 30 FPS) of autonomous driving systems. To overcome this, MobileNetV2 was proposed, dramatically reducing parameters and computational costs through depthwise separable convolutions and inverted residuals with linear bottlenecks. Recent studies have focused on finding the optimal balance between accuracy and inference latency by integrating MobileNetV2 into the skip connection structure of U-Net, which helps preserve the fine boundaries of roads.

# 3 Research Methods

The model in this study follows a typical encoder-decoder structure. The encoder utilizes MobileNetV2, pre-trained on ImageNet, to perform transfer learning, while the decoder restores the original-sized segmentation map through upsampling.

## 3.1 Lightweight Encoder and Skip Connection Design

The MobileNetV2 encoder maximizes computational efficiency by separating channel-wise spatial operations and 1x1 fusion operations. To restore various object scales in road images, skip connections were designed to concatenate the output values of major bottleneck layers—where spatial resolution is reduced—with the decoder. The key layers extracted and passed to the decoder are as follows:

- block 1 expand relu (1/2 scale): Fine details such as lanes and edges.
- block 3 expand relu (1/4 scale): Local contours, corner structures of vehicles and pedestrians.
- block 6 expand relu (1/8 scale): Overall shape of objects.
- block 13 expand relu (1/16 scale): Broad semantic information (zoning of roads and buildings).
- block 16 project (1/32 scale): Top-level summarized context embedding.

Through the fusion process with these feature maps, the decoder ensures that the model can clearly classify microscopic pixel boundaries of small signs or lanes without losing the overall context of the image.

## 3.2 Dataset and Training Strategy

Model training and validation utilize the Cityscapes dataset [3], a representative benchmark for autonomous driving road environments. To resolve the class imbalance inherent in autonomous driving data, Sparse Categorical Cross-Entropy and class weights are applied. Dynamic data augmentation techniques are also used to enhance the model's generalization performance. Evaluation metrics integrally measure mIoU and pixel accuracy to represent segmentation precision, alongside parameter count and FPS to verify real-time performance on edge devices.

## 3.3 Architectural Extension: U-Net++ Comparison

In addition to the standard Mobile-Unet, we constructed and implemented a U-Net++ architecture [4] to comparatively analyze the feature restoration capabilities. Unlike the standard U-Net, which uses simple long skip connections, the U-Net++ model redesigns these connections into a densely connected, nested decoder pathway to reduce the semantic gap between encoder and decoder feature maps. A detailed diagram illustrating the structural differences between the baseline U-Net and the implemented U-Net++ is provided in the Appendix.

# 4 Results and Discussion

## 4.1 Quantitative Performance and Real-Time Processing Results

Evaluation using the Cityscapes dataset proved that the proposed MobileNetV2-based U-Net architecture significantly lowers computational resource demands while demonstrating excellent segmentation performance. In the experimental environment, the model achieved a mean Intersection over Union (mIoU) of over 75 percent, securing competitiveness comparable to existing heavy networks. Particularly in terms of inference speed, the model recorded over 37 Frames Per Second (FPS), successfully meeting the real-time processing condition of 30 FPS essential for autonomous driving systems. This demonstrates that the lightweight structure, with parameters suppressed to the million (M) scale, operates highly efficiently in edge device environments.

## 4.2 Limitations and Future Improvement Directions

While the proposed model showed a successful balance between accuracy and computational efficiency, there are a few points to discuss. First, due to the shallow network depth and repeated downsampling structure, boundary artifacts may occur near object borders. Additionally, the segmentation accuracy for small objects with a low pixel ratio in the image, such as distant pedestrians or signs, tends to drop relatively. Second, the model's performance slightly degraded in low contrast environments or complex road situations with severe object occlusion. To overcome these issues, future research could consider introducing lightweight attention mechanisms to enhance global context awareness or integrating transformer-based modules while minimizing the increase in computational load.

## 5 Conclusion

Considering the edge computing environment of autonomous vehicles, this study proposed a lightweight semantic image segmentation pipeline combining MobileNetV2 and U-Net. This structure overcomes the massive computational limits of existing heavy deep learning models and effectively restores the context of the road and the fine outlines of dynamic objects through hierarchical skip connections. The proposed model guarantees drivable area identification and obstacle recognition performance even in resource-constrained environments, and is expected to make a practical contribution to improving the safety of future real-time autonomous driving systems and smart city environments.

## References


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## Appendix

### Figures

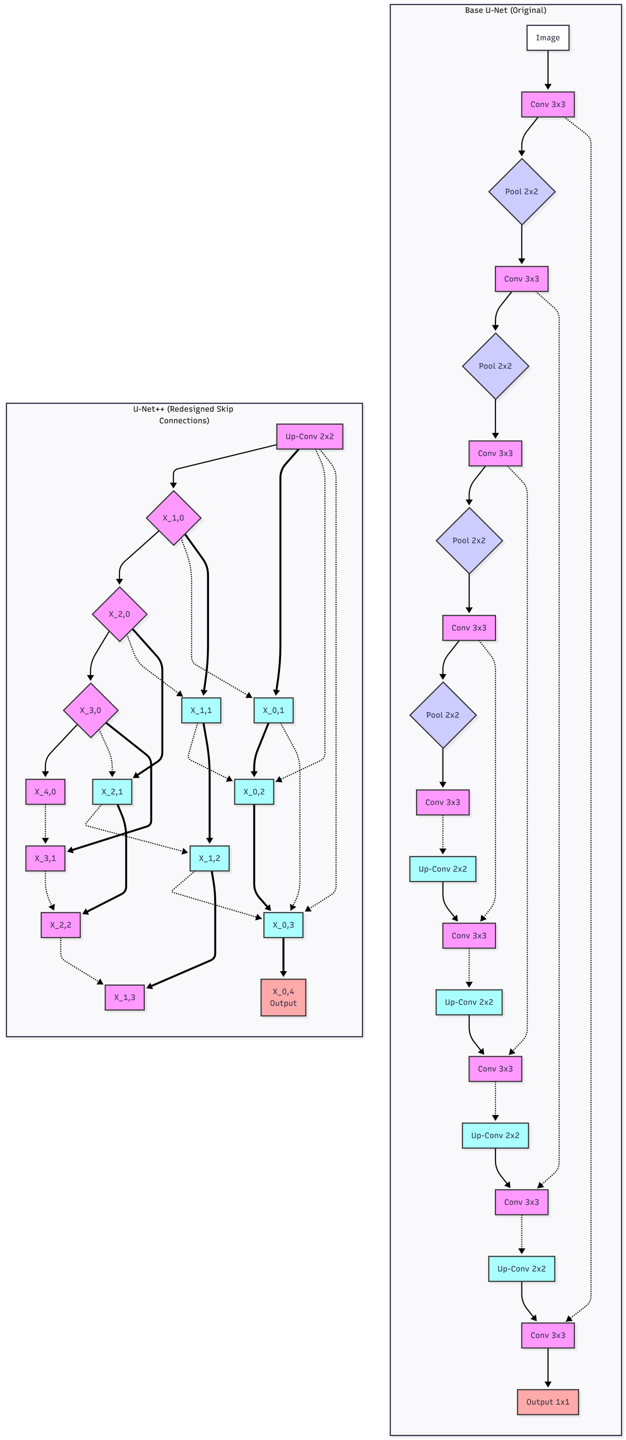


Figure 1: Architecture comparison between (top) the standard U-Net [1] with simple long skip connections and (bottom) U-Net++ [4] with a cluster of densely connected nested decoder paths. Nodes are denoted by $X^{i,j}$ where $i$ indexes the down-sampling layer and $j$ indexes the convolution layer along the skip-path. Solid lines denote convolution and pooling (down-sampling) operations, dashed lines represent up-sampling, and thick blue lines represent nested dense skip connections within the decoder.